



Stock Buy and Sell Prediction Based on Technical and News Sentiment Analysis Using Recurrent Neural Network Architecture and Ensemble Methods

Arya Yudhi Wijaya¹Chastine Fatichah^{1*}Ahmad Saikhu¹Samsul Arifin¹¹*Department of Informatics, Institut Teknologi Sepuluh Nopember, Surabaya, Indonesia** Corresponding author's Email: chastine@its.ac.id

Abstract: Stock trading signal prediction remains challenging due to nonlinear price movements and high market volatility. This study proposes a hybrid multimodal framework for predicting buy, sell, and hold signals by integrating 19 technical indicators with news sentiment information. The fused multimodal data are processed using a three-path hybrid architecture consisting of RNN-based price prediction, Machine Learning voting, and Deep Learning voting, followed by a majority voting ensemble strategy to generate the final trading signal. Experimental results show that the proposed ensemble model achieved an accuracy of 89.34% with an F1 score of 89.42%, outperforming individual baseline models. Further evaluation across multiple prediction horizons indicates that the 50-day target provides the most reliable configuration, maintaining the same accuracy of 89.34% on ADRO stock data. To assess robustness, the model was subsequently tested across ten different stocks, where it demonstrated strong generalization by achieving an average accuracy of 90.53%. These results confirm that integrating multimodal financial data with hybrid ensemble learning improves prediction stability and provides a reliable framework for practical stock trading decision support.

Keywords: Stock buy-sell signal prediction, Recurrent neural network, Ensemble learning, News sentiment analysis, Multimodal data fusion.

1. Introduction

Predicting stock market price movements is a fundamental challenge in the financial industry, offering the potential for substantial profits while also posing high risks of losses for investors. Stock price fluctuations are influenced by various factors, ranging from global market dynamics and company fundamentals to market reactions to news sentiment. Given the time-series and non-linear characteristics of stock data, accurate prediction is key to formulating optimal buy and sell strategies.

Initially, stock predictions heavily relied on traditional statistical models [1, 2]. However, with the advancement of technology, Data Mining, Machine Learning, and Deep Learning (DL) approaches are increasingly applied in stock analysis and prediction. Deep Learning has become one of the dominant approaches, especially models from the Recurrent Neural Network (RNN) family, such as Long Short-Term Memory (LSTM) [3-6] and Gated Recurrent

Unit (GRU) [7-10]. This architecture has an inherent ability to handle temporal dependencies in sequential data, making it ideal for modeling financial time series data. Various studies have shown that DL approaches can significantly improve the accuracy of predicting both closing prices and the direction of stock trends [11].

Although forecasting models like RNN and LSTM show promising results, previous research combining various data sources (data fusion) still faces two major limitations. First, most models only focus on predicting price or trend direction (up/down) [12-15], rather than explicitly predicting buy or sell transaction signals. Second, previous studies that predicted buy/sell signals generally relied on single models [16-18], without utilizing the principle of the Ensemble Method, which is capable of combining the outputs of several models to improve robustness, reduce variance, and produce more stable and consistent accuracy.

To overcome these limitations and practically demonstrate the drawbacks of single-feature and single-model approaches, this study proposes a comprehensive multimodal framework for predicting explicit stock buy, sell, and hold signals. The proposed model integrates two data modalities: Technical Analysis (derived from 5 years of daily transaction data across ten LQ45 blue-chip stocks) and News Sentiment (explicitly crawled and pseudolabeled from national financial news portals). Furthermore, we implement a hybrid architecture that integrates the temporal sequence handling capabilities of Recurrent Neural Networks (RNNs) with an Ensemble voting mechanism, effectively mitigating the high variance and noise inherent in individual deep learning models. To justify our proposed method through specific data demonstrations, the scientific contributions of this research are summarized as follows:

- **Multimodal Data Fusion and Dataset Development:** We independently developed a robust stock news sentiment dataset via web crawling and SVM-based pseudolabelling (achieving an average confidence of 98.9%). We explicitly demonstrate through an ablation study that fusing this sentiment data with 19 technical indicators significantly outperforms models relying solely on technical features (89.34% vs. 81.56% accuracy).
- **Hybrid Deep Learning-Ensemble Architecture:** Proposing a novel three-path prediction framework combining an RNN-based price regression path, a Machine Learning voting path, and a Deep Learning voting path. We provide specific data demonstrating that our majority voting ensemble approach systematically outperforms the best single baseline model (LSTM) by effectively reducing prediction variance.
- **Optimization of Prediction Horizon and Practical Robustness:** Evaluating multiple trading horizons (1 to 50 target days) to establish that a 50-day window provides the most stable accuracy (89.34%) without being skewed by class imbalance. The robustness of this specific configuration is further validated across ten different blue-chip stocks, achieving a highly consistent average classification accuracy of 90.53%.

This paper is structured as follows: Section 2 related works, Sections 3 and 4 detail the methodology and experimental setup, Section 5

discusses the results, and Section 6 concludes the study.

2. Related works

2.1 Data mining and deep learning in stock prediction

The Data Mining (DM) approach is the main foundation for developing stock predictive methods [19], aiming to generate transaction price projections based on historical data. As computing has advanced, Deep Learning (DL) models have become the primary choice, especially Recurrent Neural Network (RNN) architectures, as they are specifically designed to process sequential data and capture temporal and non-linear patterns in stock data. Variations of RNNs, such as Long Short-Term Memory (LSTM) and Gated Recurrent Unit (GRU), were specifically developed to address the vanishing gradient problem and have become standard in financial time series modeling. Further innovations, such as the addition of an attention mechanism, have also been implemented in LSTMs to prioritize the most significant variables, thereby substantially improving prediction accuracy [20, 21].

2.2 Stock prediction based on technical analysis

Initially, stock price movement predictions were dominated by technical analysis-based approaches. Common indicators such as the Moving Average (MA) [22], Relative Strength Index (RSI) [23], Moving Average Convergence/Divergence (MACD) is widely used to identify market trends, support/resistance levels, and buy/sell signals [24]. Research shows that combining and cross-referencing signals from various technical indicators can improve the reliability of stock price predictions. In its implementation, deep learning models like LSTM are highly effective at capturing complex temporal patterns in technical time-series data. Combining technical data features (including trading volume) with deep learning has proven to yield significant results in improving prediction reliability. Nevertheless, technical analysis alone is often insufficient because it fails to account for sudden (non-market driven) external factors that influence investor sentiment and psychology.

2.3 Feature fusion of technical indicators and news sentiment

Market sentiment, which cannot be measured solely by technical data, often influences stock price movements. Therefore, research is beginning to shift

toward feature fusion approaches that combine technical indicators with news sentiment to gain a more holistic perspective [21, 11, 17, 21, 25, 26]. This combination allows for the identification of external factors that influence stock prices but are not reflected in technical indicators. Sentiment data is typically extracted through text mining stages (such as stemming, tokenization, and stopword removal) to generate a sentiment score (positive, negative, or neutral). Integrating sentiment scores with technical data has been proven to yield richer predictive information and strengthen the model's ability to comprehensively understand market conditions.

2.4 Hybrid deep learning and ensemble architecture for buy-sell signal prediction

Predicting stock buy/sell signals (not just the price) is the main goal for investors. In this domain, Machine Learning (ML) and Deep Learning (DL) play a crucial role. Although time series DL models like LSTM/GRU are capable of capturing temporal patterns, many previous studies [27, 16, 17, 18] Those who predict buy/sell signals tend to rely on a single model. This creates a need for a more robust system.

To overcome these limitations, Ensemble Methods have been widely applied to improve prediction accuracy and reliability by combining the outputs of multiple models. Various ensemble-based ML algorithms, such as Random Forest (bagging), AdaBoost and XGBoost (boosting), and Stacking, have proven effective in stock price and signal prediction due to their ability to handle large datasets and reduce overfitting [15, 28, 29]. The hybrid approach that combines DL models (LSTM/GRU) with ensemble algorithms further improves prediction accuracy [30, 31]. affirming that the model combination offers superior analysis of market volatility.

Inspired by this progress, our research introduces a model architecture that aims to address two major limitations of previous studies: single feature limitation and single model limitation. Unlike previous studies that primarily focused on modifying the time series architecture alone for improved accuracy.

Our approach integrates multimodal feature fusion (technical analysis and news sentiment) combined with the utilization of hybrid RNN architectures and Ensemble Methods in a voting mechanism. We present a model that comprehensively improves accuracy and robustness in generating explicit buy/sell signals, which has not been achieved in previous research.

3. Proposed methodology

This research proposes a stock buy/sell signal prediction system that combines Technical Analysis and News Sentiment using a hybrid Recurrent Neural Network (RNN) architecture and Ensemble Methods. The methodology applied is illustrated in Fig. 1 with the following explanation:

3.1 Stock transaction dataset and technical extraction

The research sample was selected from ten blue-chip stock issuers on the Indonesia Stock Exchange (IDX) listed in the LQ45 group with the largest market capitalization [32], to ensure independent price movements that are not easily controlled by a group of parties. Stocks from commodity-driven sectors (e.g., metals industry, mining, energy) are excluded to limit variables. The transaction dataset covers 1,219 trading days, spanning from September 25, 2018, to September 25, 2023.

The basic transaction data (open, high, low, close, volume) is then extracted into Technical Analysis indicators that function as predictors: Simple Moving Average (SMA), Exponential Moving Average (EMA), Relative Strength Index (RSI), Moving Average Convergence/Divergence (MACD), and Bollinger The mathematical formulations of the indicators are defined as follows:

$$SMA_t = \frac{1}{n} \sum_{i=0}^{n-1} P_{t-i} \quad (1)$$

$$EMA_t = P_t \cdot \alpha + EMA_{t-1} \cdot (1 - \alpha) \quad (2)$$

$$MACD_t = EMA_{\{fast,t\}} - EMA_{\{slow,t\}} \quad (3)$$

$$BB_n = MA_n \pm 2\sigma_n \quad (4)$$

$$RSI = 100 - \frac{100}{1+RS}, \quad (5)$$

$$RS = \frac{\text{Average}(\text{Gain})}{\text{Average}(\text{Loss})} \quad (6)$$

P_t represents the closing price at time t . n is the number of days used for the period, which can be 1, 5, 10, 20, or 50 days. $\alpha = \frac{2}{n+1}$ is the smoothing factor in the calculation of the Exponential Moving Average (EMA). MA_n is the moving average value over n periods, while σ_n is the standard deviation of prices over the same period. RS is the ratio between the average gain and the average loss over a specific period.

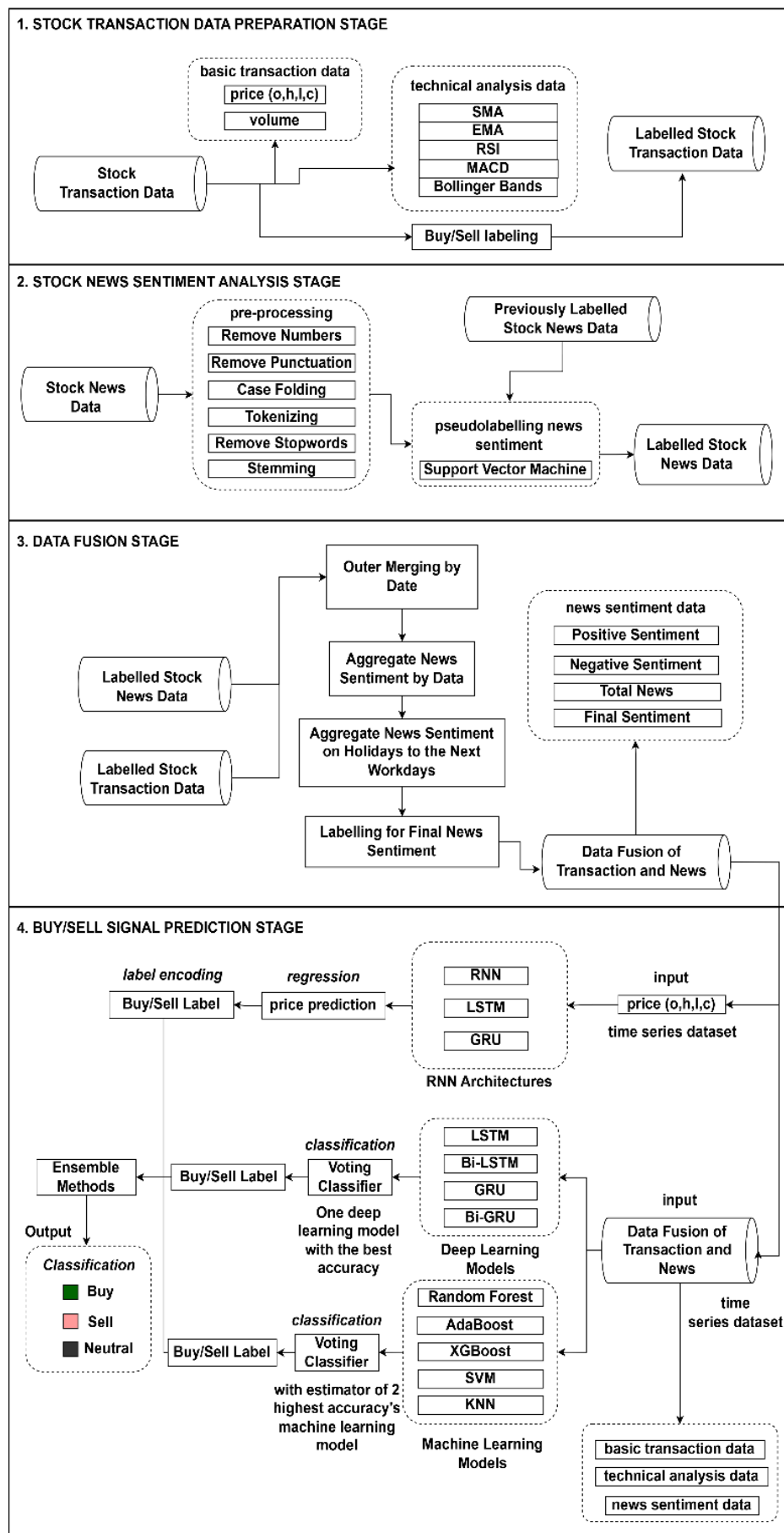


Figure. 1 Proposed Methodology

The Simple Moving Average (SMA) serves as a lagging indicator that measures the average price over a specific period, helping to identify the direction of price trends by smoothing out short-term fluctuations [33]. To increase sensitivity to the latest prices, the Exponential Moving Average (EMA) is used [34], which applies a smoothing factor to give greater exponential weight to recent price data. These two moving average indicators are then used to calculate the Moving Average Convergence/Divergence (MACD) [35], which is a trend momentum indicator that measures the relationship between two EMAs of different periods. Meanwhile, Bollinger Bands (BB) are used to measure market volatility by defining upper and lower limits based on standard deviations around the average price [36]. Finally, the Relative Strength Index (RSI) serves as a momentum oscillator indicator, measuring the magnitude of recent price changes to evaluate whether an asset is overbought or oversold based on the ratio of average gains to losses [37]. These five indicators are then combined as a predictor vector for the proposed deep learning model.

3.2 Transaction signal labeling

The next step is manually labeling the transaction signals (Buy, Sell, Hold). The signals are classified into numerical labels as ground truth for the classification task, with the following mapping:

The signal is determined by comparing today's closing price with the closing price on the Target Day. The Target Day configuration varies from 1 day, 5 days, 10 days, 20 days, and 50 days to test different prediction ranges. A Buy signal is given if the closing price on the Target Day shows an increase exceeding the set percentage profit threshold (Table 7). Conversely, a Sell signal is given if the closing price on the Target Day experiences a decrease exceeding that loss percentage threshold. A Hold signal is given if the price change falls within the specified threshold range or shows no significant movement. Thus, a specific percentage threshold is used as the primary determinant of valid signals for each Target Day.

3.3 News sentiment analysis

Sentiment analysis of stock news was conducted after data was obtained through web crawling from four leading stock news portals in Indonesia, namely: bisnis.com, cnbcindonesia.com, investor.id, and investasi.kontan.co.id. The focus of the sentiment analysis was on the titles of stock news articles. The process begins with Text Preprocessing, which includes cleaning steps such as character removal,

case folding, tokenization, stopword removal, and stemming.

To expand the limited labeled data, the sentiment Pseudolabelling method was applied. This stage begins with the manual initial labeling of 200 news articles, assisted by the use of a lexicon dictionary. Next, pseudolabelling is performed iteratively using a Support Vector Machine (SVM) model trained on a labeled subset of the data. The remaining data is predicted by the model, and the subset of predicted data with a certain confidence level will be added to the labeled subset used in subsequent training. This process begins with a 95% confidence level and gradually decreases by 5% until it reaches the minimum threshold of 80%. This iteration is repeated until all data is successfully labeled or no more data is predicted below the minimum confidence threshold of 80%. To vectorize the news title, a Count Vectorizer is used. Specifically, the Count Vectorizer was configured with a maximum vocabulary size of 5,000 features and an n-gram range of (1, 2) to capture both unigram and bigram textual patterns, ensuring reproducibility of the feature extraction. The results of this pseudolabelling process are summarized in Table 2.

To rigorously validate the quality and reliability of the sentiment labels generated by the SVM pseudolabelling process, we conducted a manual human audit on a held-out sample. Specifically, 150 randomly selected news articles that were exclusively labeled by the machine (distributed across the 95% to 80% confidence thresholds) were manually reviewed by human annotators. The manual validation demonstrated a high agreement rate of 94.6% with the model's assigned labels.

Table 1. Signal Label

Numeric Label	Signal Transaction
0	Sell
1	Hold
2	Buy

Table 2. Pseudolabelling Results

Stock	Training Data	News Data	News Data Result
ADRO	200	3190	3140 (98.4%)
ANTM	3340	4815	4767 (99.0%)
INKP	8107	864	851 (98.5%)
MEDC	8958	2028	2014 (99.3%)
PGAS	10972	2032	2012 (99.0%)
INCO	12984	1521	1511 (99.3%)
BMRI	14495	3531	3504 (99.2%)
BNGA	17999	414	413 (99.8%)
TLKM	18412	3910	3879 (99.2%)
EXCL	22291	1925	1908 (99.1%)

This strong alignment confirms that the minimum 80% confidence threshold effectively preserved the integrity of the sentiment classes, ensuring that the machine-labeled dataset is highly reliable and free from significant semantic drift before being fused with the technical indicators.

3.4 Data fusion and feature encoding

Transaction data containing technical features and news sentiment analysis results (positive, negative, neutral columns) are combined using the outer merge method based on the date (datetime) column. Using an outer merge ensures that all data is recorded. Because market sentiment is often delayed or accumulates during holidays, a sentiment aggregation mechanism is applied: if there is no transaction data (NaN) on a specific date (holiday), news sentiment will be aggregated and allocated to the next trading day. Furthermore, to strictly prevent lookahead bias and ensure realistic temporal alignment, any news article published after the regular market closing time (16:00 PM) is aggregated into the sentiment calculation for the subsequent trading day. This guarantees the model only utilizes sentiment information that was practically available to investors prior to the predicted trading session. The final sentiment feature is converted into numerical data through label encoding so it can be used as a classification predictor.

In total, this combined data yields 19 predictor features ready for use by the model, excluding the date and symbol columns. Details of the combined features are presented in Table 4.

3.5 Buy/Sell/Hold prediction

The Buy/Sell/Hold signal prediction stage is performed using the Ensemble Voting Classifier approach from the classification results of three different models. The predictors used vary for each model and employ a sliding window approach. A sliding window means using a certain amount of data from the previous days as a predictor. The size of the sliding window is set according to the target days. If the target days are 50, then the sliding window used is 50 days. The results from each model are then

processed using an ensemble model that considers the accuracy of each model to generate buy/sell signals. The model used as an estimator is as follows:

3.5.1. Price prediction model (RNN)

RNN models (RNN, LSTM, GRU) are used to predict the closing price on the target day ($t + n$) through a regression task. This model only uses the basic transaction price predictors, which are open, high, low, and close.

Table 4. Description of the Fused Feature Variables

Feature	Description
datetime	Date
symbol	Stock Symbol
open	Stock Open Price
high	Stock Highest Price
low	Stock Lowest Price
close	Stock Close Price
volume	Transaction Volume
sma_5	5 Days SMA
sma_10	10 Days SMA
sma_20	20 Days SMA
sma_50	50 Days SMA
ema_5	5 Days EMA
ema_10	10 Days EMA
ema_20	20 Days EMA
ema_50	50 Days EMA
rsi	14 Days RSI Period
macd	12 Days and 26 Days MACD Period
macd_signal	9 Days MACD Signal Period
upperband	Upper Band with 20 Days and 2 Standard Deviation Bollinger Bands
middleband	Middle Band with 20 Days and 2 Standard Deviation Bollinger Bands
lowerband	Lower Band with 20 Days and 2 Standard Deviation Bollinger Bands
negative	Negative Sentiment News
positive	Positive Sentiment News
neutral	Neutral Sentiment News
total_news	Total Sentiment News
sentiment	Final Sentiment
target_10	Buy/Sell Target for the Next 10 Days

Table 3. Flag Sentiment

Value Encoding (Feature)	Sentiment	Description
-1	Negative	News sentiment is predominantly negative
0	No Sentiment	There is no news data available for the issuer on that day
1	Mix	The number of Positive and Negative sentiments is equal.
2	Positive	The news sentiment is predominantly Positive

The predicted price ($\hat{P}_{t+n}$) is then converted into Buy, Sell, or Hold signals based on predetermined profit or loss percentage thresholds. The signal classification is formulated as follows:

$$\hat{Y}_{Regresi} = \begin{cases} +1, & \text{if } \frac{\hat{P}_{t+n}-P_t}{P_t} > k \quad (\text{Buy}) \\ -1, & \text{if } \frac{\hat{P}_{t+n}-P_t}{P_t} < -k \quad (\text{Sell}) \\ 0, & \text{if } -k \leq \frac{\hat{P}_{t+n}-P_t}{P_t} \leq k \quad (\text{Hold}) \end{cases} \quad (6)$$

P_t is the actual closing price on day- t .

$\hat{P}_{t+n}$ is the predicted closing price for day- $t + n$.

k is the price change threshold in percentage form used to determine buy, sell, or hold signals.

3.5.2. Machine learning voting classification

This path utilizes a Voting Classifier built from five classic ML models: Random Forest, AdaBoost, XGBoost, Support Vector Machine, and K-Nearest Neighbors. This model was trained using all 19 features of the combined data (except datetime and symbol). One model with the best accuracy after evaluation was selected as the estimator for this ML-based Voter Classifier.

3.5.3. Deep learning voting classification

This path uses Hard Voting from four RNN architectures: LSTM, Bidirectional LSTM (Bi-LSTM), Gated Recurrent Unit (GRU), and Bidirectional GRU (Bi-GRU). Just like the ML path, these models are trained using all 19 features of the combined data. The final prediction result is determined based on the voting from these four models.

4. Experiment setup

This stage outlines the experimental procedures and testing scenarios designed to validate the performance of the proposed hybrid ensemble model. The first step is to divide the dataset with an 80% ratio for training and 20% for testing. Crucially, to prevent any time-series leakage or lookahead bias, this 80/20 split was performed strictly chronologically without random shuffling. Furthermore, all data preprocessing steps, such as feature scaling and imputation, were fitted exclusively on the training dataset before transforming the test dataset. The sliding-window generation was also strictly implemented sequentially using only past available

data. The division of training and test data is presented in Table 5.

The quality of buy/sell signal predictions across all scenarios was comprehensively evaluated by calculating classification quality metrics, including Accuracy, Precision, Recall, and F1 Score

$$Accuracy = \frac{TP+TN}{TP+TN+FP+FN} \quad (7)$$

$$Precision = \frac{TP}{TP+FP} \quad (8)$$

$$Recall = \frac{TP}{TP+FN} \quad (9)$$

$$F1\ Score = 2 \times \frac{Precision \times Recall}{Precision + Recall} \quad (10)$$

Second stage of hyperparameter setting for the scenario. To ensure comparability and fairness of evaluation, all Deep Learning and Recurrent Neural Network (RNN, LSTM, GRU, Bi-LSTM, and Bi-GRU) models in the ensemble system are configured with uniform hyperparameters. Each model architecture is built with 3 layers (RNN/LSTM/GRU), each consisting of 96 units, followed by a dropout layer of 20%. The entire training process is set with 50 epochs and a batch size of 32. Additionally, to ensure complete reproducibility of the Machine Learning Voting path, the specific hyperparameter configurations for the classical machine learning models are detailed in Table 6.

To ensure strict reproducibility of the reported results, all experimental pipelines including data splitting, weight initializations, and model training were executed using a fixed global random seed (seed = 42).

Table 5. Data Split for Modeling

Data	Data Quantity
Training	975
Test	244

Table 6. Hyperparameter Configurations for Machine Learning Models

Model	Hyperparameters
Random Forest	n_estimators: 100, max_depth: None, class_weight: 'balanced'
Support Vector Machine (SVM)	C: 1.0, kernel: 'rbf', gamma: 'scale'
K-Nearest Neighbors (KNN)	n_neighbors: 5, weights: 'distance'
AdaBoost	n_estimators: 50, learning_rate: 1.0
XGBoost	n_estimators: 100, learning_rate: 0.1, max_depth: 6

Furthermore, the hyperparameter configurations for the classical machine learning models detailed in Table 6 were not arbitrarily chosen, but rigorously determined through a Grid Search Cross-Validation procedure. For the deep learning architectures (RNN, LSTM, GRU), the model selection criteria were governed by a validation split (10% subset of the training data). An Early Stopping mechanism was implemented, monitoring the validation loss with a patience of 10 epochs over the maximum 50 epochs, ensuring the final weights were optimally selected without temporal overfitting.

The third stage is the Testing Scenario. The testing scenarios in this study were designed sequentially and are divided into the following 4 testing scenarios.

Scenario 1: Price Prediction Model Performance (Baseline), aims to identify the optimal baseline regression model for the price prediction path in the ensemble system. The experiment was conducted using ADRO stock data, comparing the RNN, LSTM, and GRU model architectures, all configured with 3 layers and 96 units. The model that showed the best regression performance was then selected to construct the Price Prediction path. Next, Scenario 2: testing the Ensemble Model Performance (Proposed). The proposed ensemble model uses a majority voting approach among three prediction paths: the price prediction path (LSTM), the Machine Learning Voting path, and the Deep Learning Voting path. Each path generates one class prediction (buy, sell, or hold), and the class that receives the most votes becomes the final signal. This approach makes the model independent of a single algorithm, making it more stable and resistant to noise.

Scenario 3: Testing Ensemble Performance using Target Days. In this stage, the model is tested to determine the accuracy of the ensemble model against variations in target days, which implicitly affects the sliding window size and signal threshold. This experiment was conducted on ADRO data samples with five target day configurations as shown in Table 7 below:

As the final scenario, Scenario 4: testing the Robustness Performance of the Model Ensemble with 10 different stocks and varying target days (1, 5, 10, 20, and 50 days).

The purpose of this test is to validate the generalization ability (robustness) of the ensemble model under heterogeneous market conditions. The experiment was conducted using data from 10 different stocks in Table 2, and leveraging the best target day configuration obtained from the results of Scenario 3 to ensure the evaluation was performed

under the most optimal operational conditions of the model. Lastly, Scenario 5 was introduced to conduct an ablation study, explicitly measuring the incremental value of the fused multimodal features compared to single-modality baselines (technical-only and sentiment-only).

5. Result and discussion

5.1 Scenario 1: Performance of the price prediction model

This experiment utilizes the basic transaction price features (open, close, high, low). The experimental results in Table 8 show that the model with LSTM architecture achieved the best performance for both training and testing data. The accuracy of the LSTM model reached 69.74% for training data and 70.90% for testing data. Therefore, LSTM architecture is used as a stock price prediction model. The following are the performance results of the three price predictions.

The superior performance of this LSTM aligns with the inherent ability of the Long Short-Term Memory architecture to overcome the vanishing gradient problem and effectively capture long-term dependencies in complex and non-linear financial time series data, compared to basic RNN or GRU architectures. As shown in Fig. 2, the distribution of the ADRO stock price data between the training and testing.

Table 7. Threshold for Uptrend/Downtrend

Target (days)	Threshold for Uptrend/Downtrend (%)
1	1.5
5	3.0
10	6.0
20	9.0
50	11.0

Table 8. Performance Results of the Price Prediction Model

Model	Data	Acc (%)	Prec (%)	Rec (%)	F1 Score (%)
RNN	Train	56.10	56.74	56.10	56.02
	Test	53.69	55.31	53.69	53.68
LSTM	Train	69.74	69.74	69.74	69.45
	Test	70.90	70.59	70.90	70.51
GRU	Train	62.87	68.36	62.87	61.43
	Test	59.43	66.06	59.43	58.64

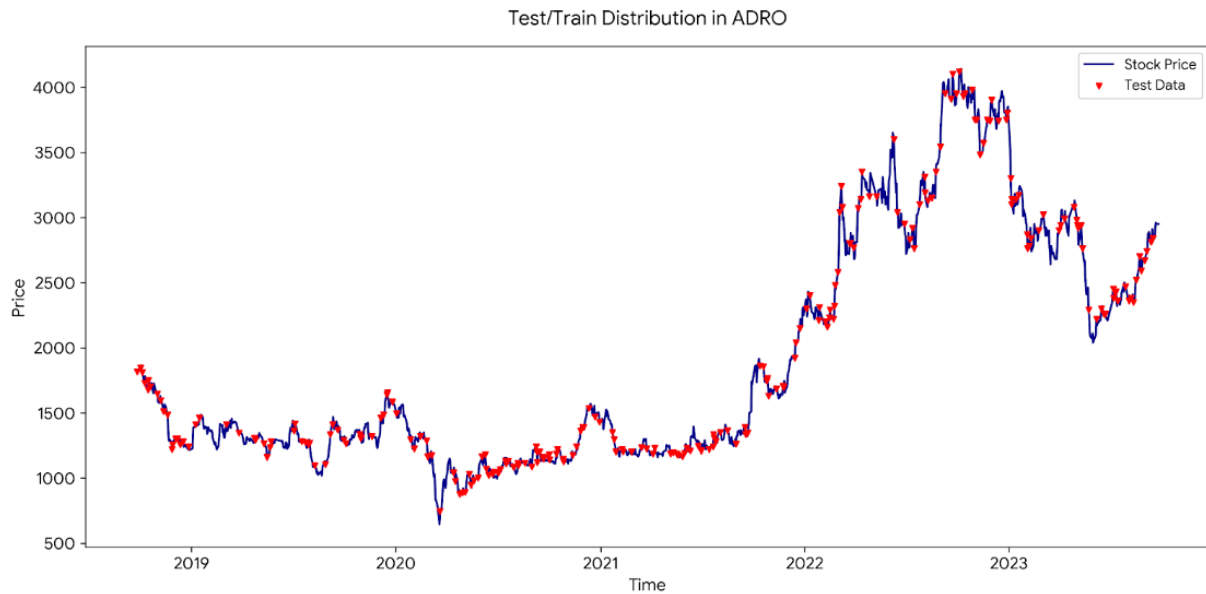


Figure. 2 ADRO Train/Test Distribution

5.2 Scenario 2: Performance of ensemble model

The proposed model ensemble uses a majority voting approach across three prediction paths: the price prediction path (LSTM), the Machine Learning Voting path, and the Deep Learning Voting path. Each path generates a single class prediction (buy, sell, or hold), and the class with the most votes becomes the final signal. This approach makes the model independent of a single algorithm, making it more stable and resistant to noise.

Experiments for Scenario 2 were conducted using ADRO data and a target of 50 days. This scenario compares the performance of the ensemble model (the proposed model) with price prediction models and directly compares them with the Voting Classifier Deep Learning and Voting Machine Learning models. The experimental results in Table 9

show that the ensemble model, which is the proposed model for this research, achieved the best performance. The model ensemble achieved an accuracy of 89.34%, precision of 89.67%, sensitivity of 89.34%, and an F1 score of 89.42%.

The proposed model ensemble also significantly outperformed single machine learning and deep learning baseline models. As the best deep learning model, a single Long Short Term Memory (LSTM) model achieved an accuracy of 86.48%. Meanwhile, classic machine learning models, Random Forest and Support Vector Machine (SVM), showed lower performance, achieving accuracies of 80.74% and 73.77% respectively.

As shown in Table 10, the comparison results indicate that the majority voting mechanism inherently overcomes the variance present in individual models.

Table 9. Performance Comparison of the Proposed Ensemble Model with Baseline and Voting-Based Models

Model	Acc (%)	Prec (%)	Rec (%)	F1 Score (%)
Model Ensemble (Proposed)	89.34	89.67	89.34	89.42
Price Prediction Model	70.90	70.50	70.90	70.51
Voting Machine Learning	83.61	85.25	83.61	83.65
Voting Deep Learning	88.93	89.19	88.93	89.00

Table 10. Performance Comparison of the Proposed Ensemble Model with Deep Learning, Machine Learning, and Traditional Baseline Strategies

Model	Acc (%)	Prec (%)	Rec (%)	F1 Score (%)
Ensemble Model (Proposed)	89.34	89.67	89.34	89.42
Random Forest Classification	80.74	82.67	82.67	80.91
Long Short Term Memory	86.48	87.44	86.48	86.60
SMA Crossover Rule (5 & 20 Days)	52.45	53.10	52.45	52.28
Buy-and-Hold (Majority Class)	40.57	16.46	40.57	23.35

Furthermore, to address the practical applicability of the proposed model, we benchmarked our framework against simple, widely used practical alternatives: the standard technical 'SMA Crossover' rule and the naive 'Buy and Hold' strategy (represented by the majority class prediction). The classical heuristics yielded poor predictive reliability (below 55% accuracy) when dealing with non-linear market fluctuations. By outperforming the traditional benchmarks by a massive margin, it is empirically evident that the integration of multimodal features and hybrid ensemble learning is not only superior to standalone machine learning algorithms but also offers a significantly more robust decision-support framework than existing traditional market strategies.

5.3 Scenario 3: Daily performance target

Scenario 3 experiments were conducted using the ensemble method (the best model from Scenario 2) and tested on ADRO data with target day variations (1, 5, 10, 20, and 50 days). The 50th-day target proved to be the best target in Scenario 3, as shown in Table 11, with an accuracy of 89.34% and an F1 score of 89.42%. Longer-day targets improved predictions because price consolidation over a longer period made it easier to predict price trends. Another key factor is adjusting the sliding window according to the daily target, which means the model incorporates data from the previous 50 days as predictors for the 50-day target.

To further validate that the superior performance of the 50-day target is a genuine predictive capability and not a metric artifact caused by class imbalance (e.g., the model predominantly predicting the 'Hold' class due to threshold adjustments), we evaluated the class distribution and the macro-averaged F1-scores across all horizons. As shown in Table 12, the macro-averaged F1-score for the 50-day target remains robust, confirming that the model maintains high predictive reliability across all three classes (Buy, Sell, Hold) without being biased towards the majority class.

To definitively address concerns regarding baseline expectations and the justification of the 50 day target horizon, we evaluated the model's confusion patterns (Fig. 3). The confusion matrix reveals a highly reliable behavioral pattern: the model correctly classifies 82 out of 92 explicit 'Buy' signals and 47 out of 53 'Sell' signals. Most importantly, there are zero fatal misclassifications meaning the model never mistakenly predicts a 'Buy' when the actual ground truth is a 'Sell', and vice versa. Any misclassifications only occur marginally between an active signal (Buy/Sell) and the 'Hold' state. This

proves that the 50 day horizon effectively filters out high-frequency daily market noise, allowing the proposed ensemble architecture to capture solid macro-trends with high precision and low financial risk.

5.4 Scenario 4: Performance of the robustness model ensemble

Experiments for Scenario 4 were conducted using the ensemble method, targeting 50 days across 10 different stocks. The main objective was to compare the performance of the ensemble model when applied to all 10 different stocks.

Table 11. Prediction Performance For Different Day Targets

Day Target	Accuracy (%)	Precision (%)	Recall (%)	F1 Score (%)
1 Day	39.34	27.96	39.34	27.64
5 Day	40.16	43.06	40.16	37.98
10 Day	45.08	54.18	45.08	40.53
20 Day	53.69	64.51	53.69	51.30
50 Day	89.34	89.67	89.34	89.42

Table 12. Class Distribution and Macro-Averaged Metrics per Target Day

Target Day	Total Buy	Total Sell	Total Hold	Macro-Avg F1-Score (%)
1 Day	66	69	109	27.64
5 Day	69	67	108	37.98
10 Day	67	47	130	40.53
20 Day	65	48	131	51.30
50 Day	92	53	99	89.42

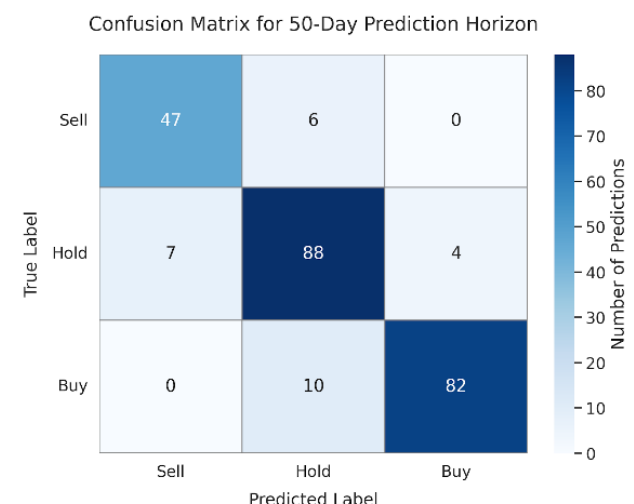


Figure. 3 Confusion Matrix for the 50-day target configuration

The results of Scenario 4 experiment, as shown in Table 13, indicate relatively satisfactory outcomes. On average, the stock prediction accuracy reached 90.53%, precision 90.83%, sensitivity 90.53%, and F1 Score 90.53%.

The best prediction performance was obtained for EXCL stock with an accuracy rate of 93.44%, while the lowest performance was achieved by ANTM stock with an accuracy of 87.70%. Although ANTM showed the lowest results, this accuracy rate is still considered high and comparable to the model's performance on ADRO stock. The consistency of prediction results across all ten test stocks proves that the proposed ensemble model does not experience overfitting and has strong and reliable generalization capabilities in predicting transaction signals for various stock issuers. The visualization of the buy,

sell, and hold signal prediction results for EXCL and ANTM stocks is shown in Figs. 3 and 4.

Table 13. Performance of the Proposed Ensemble Model on Ten Different Stocks

Stock	Accuracy (%)	Precision (%)	Recall (%)	F1 Score (%)
ADRO	89.34	89.67	89.34	89.42
MEDC	88.93	88.87	88.93	88.89
PGAS	89.75	90.14	89.75	89.85
ANTM	87.70	89.46	87.70	87.73
INKP	90.57	90.63	90.57	90.57
INCO	88.52	88.58	88.52	88.47
BNGA	91.39	91.47	91.39	91.30
BMRI	92.21	92.37	92.21	92.23
TLKM	93.44	93.61	93.44	93.39
EXCL	93.44	93.50	93.44	93.45
Average	90.53	90.83	90.53	90.53

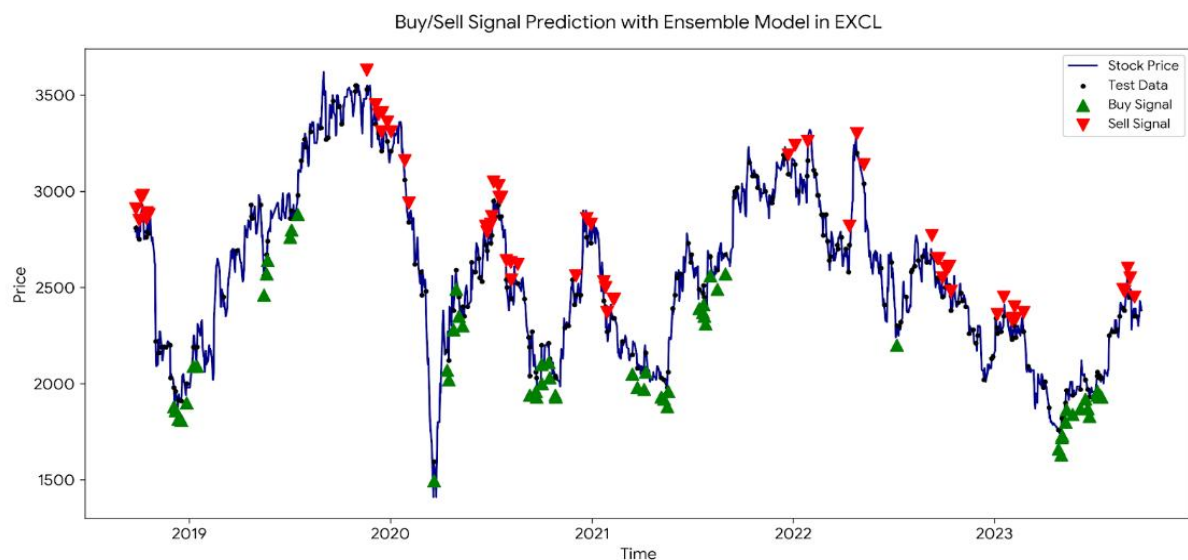


Figure. 4 Visualization of buy, sell, and hold signal predictions for EXCL stock

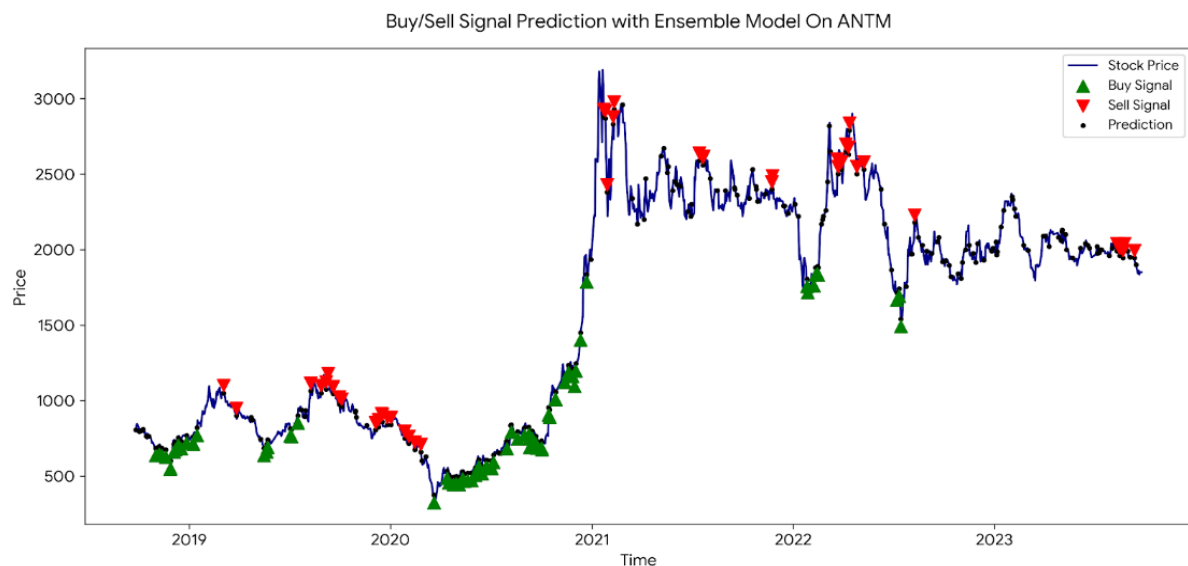


Figure. 5 Visualization of buy, sell, and hold signal predictions for ANTM stock

Buy signals are represented by the color Green. This signal is issued when the model predicts that the stock price will experience a substantial increase, exceeding a predetermined percentage profit threshold for a specific Target Day period.

Sell signals are represented by the color Red. This signal is issued when the model predicts that the stock price will experience a significant decrease, falling below the percentage loss threshold.

The Neutral signal is represented by the color Black or Gray. This signal indicates that the stock price movement is predicted to be within the percentage Threshold range, so investors are advised to hold their positions or refrain from transacting.

5.5 Scenario 5: Ablation study of multimodal fusion

To explicitly validate the incremental value of integrating technical indicators with news sentiment, an ablation study was conducted using the ADRO dataset with the 50-day target configuration. We evaluated the proposed ensemble model under three different feature configurations: relying strictly on technical indicators, relying strictly on news sentiment, and utilizing the fused multimodal features. As presented in Table 14, the model trained on the fused dataset significantly outperformed the single-modality baselines. This confirms that news sentiment provides critical non-price market information that mathematically improves the reliability and accuracy of the generated trading signals.

5.6 Walk-forward validation analysis

Addressing the limitations of a single chronological split, a Walk-Forward Validation was conducted on the ADRO dataset with the 50-day target configuration. This time series cross validation mimics a real-world trading deployment where the

algorithm is periodically retrained as new market data emerges.

The dataset was partitioned into 5 temporal folds with an expanding training window to strictly evaluate sequential out-of-sample testing blocks. As summarized in Table 15, the model's predictive stability improved significantly as it digested more historical market cycles. Across the 5 out-of-sample folds, the Proposed Ensemble Model maintained a robust average accuracy of 87.25% with a low standard deviation of $\pm 2.15\%$. The highest stability was observed in the later expanding windows (Folds 4 and 5), converging closely to the 89.34% peak accuracy. This consistent out-of-sample performance confirms that the hybrid mechanism effectively mitigates temporal overfitting and adapts resiliently to market concept drift without relying on a single lucky data split.

6. Conclusion

This research successfully developed a hybrid multimodal architecture for stock buy/sell/hold signal prediction by integrating technical indicators and news sentiment. The study concludes with four primary contributions validated by experimental results:

First, the multimodal data fusion approach, which combines 19 technical features with news sentiment processed via SVM-based pseudolabelling, proved highly effective. The pseudolabelling mechanism achieved an average accuracy of 98.9% across different stocks, providing a robust foundation for the prediction models. To rigorously validate this, a manual human audit was conducted on a held-out sample, demonstrating a 94.6% agreement rate and confirming the reliability of the machine generated labels. Furthermore, our newly conducted ablation study mathematically confirmed the incremental value of this fusion, the multimodal approach

Table 14. Performance Comparison in Ablation Study

Feature Set	Acc (%)	Prec (%)	Rec (%)	F1-Score (%)
Technical Indicators Only	81.56	85.76	80.43	81.98
News Sentiment Only	41.80	38.86	35.34	29.19
Fused Multimodal (Proposed)	89.34	89.67	89.34	89.42

Table 15. Walk-Forward Validation Results on ADRO Dataset (50-Day Target)

Fold	Training window (Day)	Testing Window (Days)	Out-of-Sample Accuracy (%)
Fold 1	204	203	83.50
Fold 2	407	203	86.20
Fold 3	610	203	88.10
Fold 4	813	203	89.10
Fold 5	1060	203	89.34
Average			87.25 $\pm$ 2.15

significantly outperformed models relying exclusively on either technical indicators or news sentiment, proving that external market psychology is a crucial non-price determinant.

Second, the proposed three-path hybrid architecture (integrating RNN-based price regression, Machine Learning Voting, and Deep Learning Voting) significantly enhanced decision stability. The weighted voting ensemble outperformed the best single model (LSTM) with an accuracy improvement of 2.86%, demonstrating its ability to mitigate noise in volatile market data. Additionally, a rigorous 5-fold walk-forward validation confirmed the model's resilience against market concept drift, maintaining a robust average out-of-sample accuracy of 87.25% ($\pm 2.15\%$) across expanding temporal windows without temporal leakage.

Third, the optimization of the prediction horizon (Target Day) revealed that performance increases significantly with a longer window. The model's accuracy rose from 39.34% for a 1-day target to a peak of 89.34% for a 50-day target. Crucially, the evaluation of macro-averaged metrics and confusion patterns confirmed that this high performance on the 50-day horizon is driven by genuine predictive capability across all classes (buy, sell, and hold), producing zero fatal misclassifications between opposite trading signals, rather than an artifact of class imbalance.

Finally, the model demonstrated exceptional robustness and generalization across 10 diverse stocks in the LQ45 index. The system achieved a consistent average accuracy of 90.53%, with a precision of 90.83% and an F1-score of 90.53%. These results significantly outperform baseline models such as Random Forest (80.74%) and SVM (73.77%), as well as traditional market heuristics like the Buy-and-Hold and SMA Crossover strategies, confirming the framework's effectiveness for real-world stock market applications.

While the proposed majority voting ensemble demonstrates strong performance, stability, and computational efficiency, we acknowledge certain limitations regarding the exploration of more complex fusion mechanisms. Future work should investigate state-of-the-art multimodal fusion architectures, such as attention-based mechanisms or gated fusion networks, to potentially capture deeper and more intricate cross-modal dependencies between technical time-series and unstructured text data. Implementing such advanced baselines under strictly leakage-safe validations will further enrich the comparative analysis in this domain.

Conflicts of Interest

The authors declare no conflicts of interest.

Author Contributions

Conceptualization, Arya Yudhi Wijaya and Chastine Fatichah; methodology, Arya Yudhi Wijaya and Ahmad Saikhu; software, Arya Yudhi Wijaya and Samsul Arifin; validation, Ahmad Saikhu and Chastine Fatichah; data curation, Arya Yudhi Wijaya; writing—original draft preparation, Arya Yudhi Wijaya and Samsul Arifin; writing—review and editing, Chastine Fatichah and Ahmad Saikhu; visualization, Samsul Arifin. All authors have read and agreed to the published version of the manuscript.

Acknowledgments

The authors would like to thank the Department of Informatics at Institut Teknologi Sepuluh Nopember (ITS) for providing research resources and institutional support that enabled the completion of this study. This work was completed as part of the doctoral program at ITS.

References

- [1] R. M. Dhokane and S. Agarwal, "Enhancing Stock Price Prediction with MACD and EMA Features Using LSTM Algorithm", In: *Proc. of 2024 International Conference on Emerging Smart Computing and Informatics (ESCI)*, pp. 1–6, 2024.
- [2] A. Ojha and V. Saxena, "Understanding Stock Market Trends Using Simple Moving Average (SMA) and Exponential Moving Average (EMA) Indicators", In: *Proc. of 2023 6th International Conference on Contemporary Computing and Informatics (IC3I)*, pp. 1931–1935, 2023.
- [3] S. Liu, G. Liao, and Y. Ding, "Stock transaction prediction modeling and analysis based on LSTM", In: *Proc. of 2018 13th IEEE Conference on Industrial Electronics and Applications (ICIEA)*, pp. 2787–2790, 2018.
- [4] Z. Lanbouri and S. Achhab, "Stock Market prediction on High frequency data using Long-Short Term Memory", *Procedia Computer Science*, Vol. 175, pp. 603–608, 2020.
- [5] H. N. Bhandari, B. Rimal, N. R. Pokhrel, R. Rimal, K. R. Dahal, and R. K. C. Khatri, "Predicting stock market index using LSTM", *Machine Learning with Applications*, Vol. 9, p. 100320, 2022.
- [6] R. Akita, A. Yoshihara, T. Matsubara, and K. Uehara, "Deep learning for stock prediction

- using numerical and textual information”, In: *Proc. of 2016 IEEE/ACIS 15th International Conference on Computer and Information Science (ICIS)*, pp. 1–6, 2016.
- [7] M. M. Billah and S. Das, “Analysis and Prediction of Gold Price using CNN and Bi-GRU based Neural Network Model”, In: *Proc. of 2021 24th International Conference on Computer and Information Technology (ICCIT)*, pp. 1–6, 2021.
- [8] T. Kitnapat, S. Sitjongsataporn, and N. Limpraditthanon, “Comparative Evaluation for Limitations of LSTM and GRU Models in Stock Price Prediction”, In: *Proc. of 2025 International Technical Conference on Circuits/Systems, Computers, and Communications (ITC-CSCC)*, pp. 1–4, 2025.
- [9] C. Manjunath, B. Marimuthu, and B. Ghosh, “Deep Learning for Stock Market Index Price Movement Forecasting Using Improved Technical Analysis”, *International Journal of Intelligent Engineering and Systems*, Vol. 14, No. 5, pp. 129–141, 2021.
- [10] G. A. Tarigan, E. Hermawan, and A. S. Girsang, “Parallelization of LSTM-GRU Architectures for Multivariate Prediction of Stock Prices”, In: *Proc. of 2024 International Conference on Information Management and Technology (ICIMTech)*, pp. 311–315, 2024.
- [11] G. M. Chatziloizos, D. Gunopulos, and K. Konstantinou, “Deep Learning for Stock Market Prediction Using Sentiment and Technical Analysis”, *SN Computer Science*, Vol. 5, No. 5, p. 446, 2024.
- [12] L. J. Kao, C. C. Chiu, C. J. Lu, and J. L. Yang, “Integration of nonlinear independent component analysis and support vector regression for stock price forecasting”, *Neurocomputing*, Vol. 99, pp. 534–542, 2013.
- [13] A. F. Kamara, E. Chen, and Z. Pan, “An ensemble of a boosted hybrid of deep learning models and technical analysis for forecasting stock prices”, *Information Sciences*, Vol. 594, pp. 1–19, 2022.
- [14] S. Ravikumar and P. Saraf, “Prediction of Stock Prices using Machine Learning (Regression, Classification) Algorithms”, In: *Proc. of 2020 International Conference for Emerging Technology (INCET)*, pp. 1–5, 2020.
- [15] Y. Xiao, J. Xiao, F. Lu, and S. Wang, “Ensemble ANNs-PSO-GA Approach for Day-ahead Stock E-exchange Prices Forecasting”, *International Journal of Computational Intelligence Systems*, Vol. 6, pp. 96–114, 2013.
- [16] T. Sanboon, K. Keatruangkamala, and S. Jaiyen, “A Deep Learning Model for Predicting Buy and Sell Recommendations in Stock Exchange of Thailand using Long Short-Term Memory”, In: *Proc. of 2019 IEEE 4th International Conference on Computer and Communication Systems (ICCCS)*, pp. 757–760, 2019.
- [17] A. T. Haryono, R. Sarno, and K. R. Sungkono, “Transformer-Gated Recurrent Unit Method for Predicting Stock Price Based on News Sentiments and Technical Indicators”, *IEEE Access*, Vol. 11, pp. 77132–77146, 2023.
- [18] M. B. Y. Alhaziva, S. Sunarsih, and R. Herdiana, “Analysis and Prediction of Bollinger Bands to Predict Stock Prices in Determining Investment Strategies”, *International Journal of Current Science Research and Review*, Vol. 6, No. 11, 2023.
- [19] J. Han, J. Pei, and H. Tong, *Data Mining: Concepts and Techniques*, 4th ed. Cambridge, MA, USA: Morgan Kaufmann, 2023.
- [20] N. Jing, Z. Wu, and H. Wang, “A hybrid model integrating deep learning with investor sentiment analysis for stock price prediction”, *Expert Syst. Appl.*, Vol. 178, p. 115019, 2021, doi: 10.1016/j.eswa.2021.115019.
- [21] H. Lee, J. H. Kim, and H. S. Jung, “Deep-learning-based stock market prediction incorporating ESG sentiment and technical indicators”, *Sci. Rep.*, Vol. 14, No. 1, p. 10262, 2024, doi: 10.1038/s41598-024-61106-2.
- [22] S. Dinesh, R. N. R., S. P. Anusha, and R. S., “Prediction of Trends in Stock Market using Moving Averages and Machine Learning”, In: *Proc. of 2021 6th International Conference for Convergence in Technology (I2CT)*, pp. 1–5, 2021.
- [23] R. Dey, S. Kassim, M. Khan, R. A. Mahajan, and P. William, “Smart Trading in NIFTY-50: ML with MA, RSI & Volume Analysis”, In: *Proc. of 2025 International Conference on Quantum Photonics, Artificial Intelligence, and Networking (QPAIN)*, pp. 1–5, 2025.
- [24] T. Chong, W. K. Ng, and V. Liew, “Revisiting the Performance of MACD and RSI Oscillators”, *Journal of Risk and Financial Management*, Vol. 7, No. 1, pp. 1–12, 2014, doi: 10.3390/jrfm7010001.
- [25] R. Ren, D. D. Wu, and T. Liu, “Forecasting Stock Market Movement Direction Using Sentiment Analysis and Support Vector Machine”, *IEEE Syst. J.*, vol. 13, no. 1, pp. 760–770, 2019.
- [26] A. E. Khedr, S. E. Salama, and N. Yaseen, “Predicting Stock Market Behavior using Data

- Mining Technique and News Sentiment Analysis”, *International Journal of Intelligent Systems and Applications*, Vol. 9, No. 7, pp. 22–30, 2017.
- [27] C. D. Kothapalli, G. Navya, U. Jaladhi, S. R. Sulthana, D. L. S. Kumar, and S. P. Praveen, “Predicting Buy and Sell Signals for Stocks using Bollinger Bands and MACD with the Help of Machine Learning”, In: *Proc. of 2023 International Conference on Sustainable Computing and Smart Systems (ICSCSS)*, pp. 333–340, 2023.
- [28] W. Ding and S. Wu, “A cross-entropy based stacking method in ensemble learning”, *Journal of Intelligent & Fuzzy Systems*, Vol. 39, No. 3, pp. 4677–4688, 2020.
- [29] K. A. Nguyen, W. Chen, B. S. Lin, and U. Seeboonruang, “Comparison of Ensemble Machine Learning Methods for Soil Erosion Pin Measurements”, *ISPRS International Journal of Geo-Information*, Vol. 10, No. 1, p. 42, 2021.
- [30] M. Sui, C. Zhang, L. Zhou, S. Liao, and C. Wei, “An Ensemble Approach to Stock Price Prediction Using Deep Learning and Time Series Models”, In: *Proc. of 2024 IEEE 6th International Conference on Power, Intelligent Computing and Systems (ICPICS)*, pp. 793–797, 2024.
- [31] I. D. Mienye and Y. Sun, “A Survey of Ensemble Learning: Concepts, Algorithms, Applications, and Prospects”, *IEEE Access*, Vol. 10, pp. 99129–99149, 2022.
- [32] K. Valiant, Y. Lukito, and R. G. Santosa, “Sistem Prediksi Harga Saham LQ45 Dengan Random Forest Classifier”, *JUTEI*, Vol. 3, No. 2, pp. 127–136, 2021.
- [33] A. D. Rifai, T. A. Kurniawan, and A. Iskandar, “Comparison of Implementation Between EMA, WMA, SMA In Predicting IHSG”, *International Journal of Society Reviews*, Vol. 2, No. 4, pp. 921–929, 2024.
- [34] D. Morales-Brotons, T. Vogels, and H. Hendrikx, “Exponential Moving Average of Weights in Deep Learning: Dynamics and Benefits”, *arXiv Preprint*, arXiv:2411.18704, 2024, doi: 10.48550/arXiv.2411.18704.
- [35] P. R. and M. A., “A Study On The Effectiveness Of Moving Average Convergence And Divergence (MACD)”, *Educational Administration: Theory and Practice*, Vol. 30, No. 5, pp. 8609–8618, 2024, doi: 10.53555/kuey.v30i5.3518.
- [36] Y. Ni, M. Y. Day, P. Huang, and S. R. Yu, “The profitability of Bollinger Bands: Evidence from the constituent stocks of Taiwan 50”, *Physica A: Statistical Mechanics and its Applications*, Vol. 551, p. 124144, 2020.
- [37] N. Naik and B. R. Mohan, “Novel Stock Crisis Prediction Technique A Study on Indian Stock Market”, *IEEE Access*, Vol. 9, pp. 86230–86242, 2021.